

Trabajo Practico Final

Reproductor MP3

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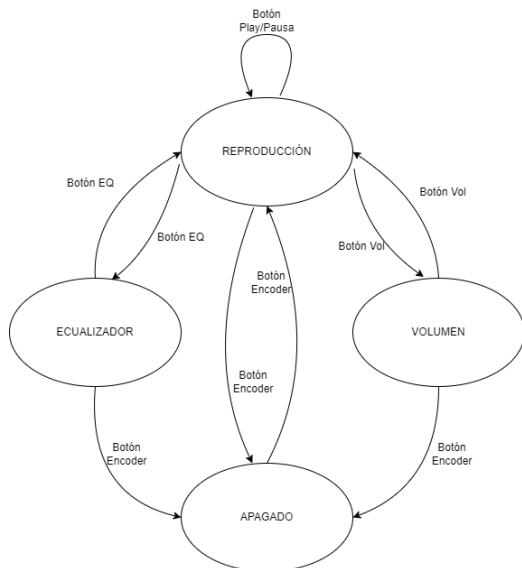
Santiago Feldman

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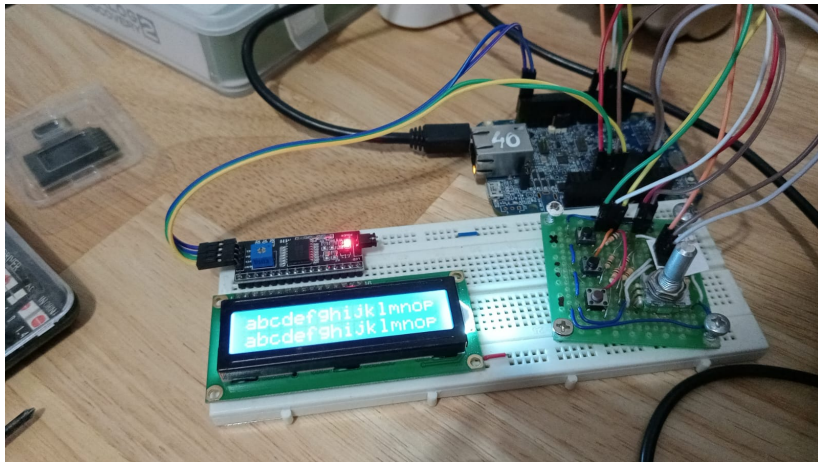
Febrero 2024

Veámoslo en funcionamiento!!

Máquina de estados



Interfaz física



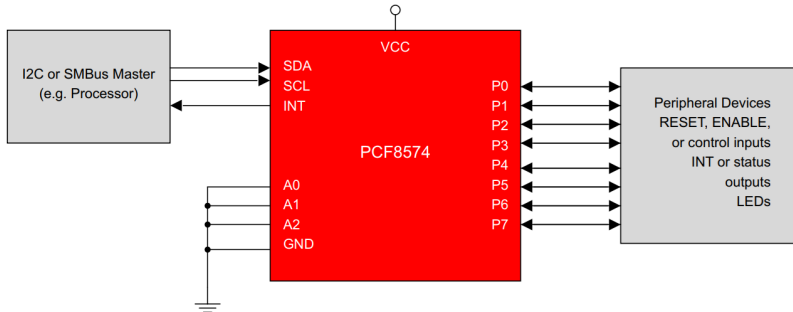
Display 16x2



PCF8574

SCPS068J – JULY 2001 – REVISED MARCH 2015

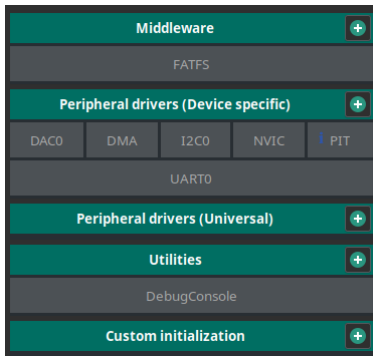
PCF8574 Remote 8-Bit I/O Expander for I²C Bus



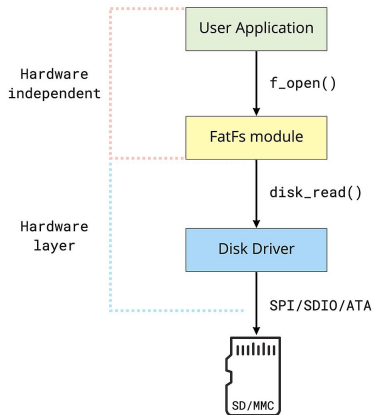
Display 16x2

```
1 void displaySendPeriodicISR(){
2
3     static int sendingState = 0;
4     static uint8_t wordSend;
5
6     switch(sendingState){
7     case 0:
8         if (getFillLevel(&displayDataBuffer) != 0){
9             wordSend = getNext(&displayDataBuffer);
10            WriteDriverByteIIC(&wordSend);
11            sendingState=1;
12        }
13        break;
14     case 1:
15         wordSend |= En;
16         WriteDriverByteIIC(&wordSend);
17         sendingState=2;
18        break;
19     case 2:
20         wordSend &= ~En;
21         WriteDriverByteIIC(&wordSend);
22         sendingState = 0;
23        break;
24     default:
25
26        break;
27    }
28 }
```

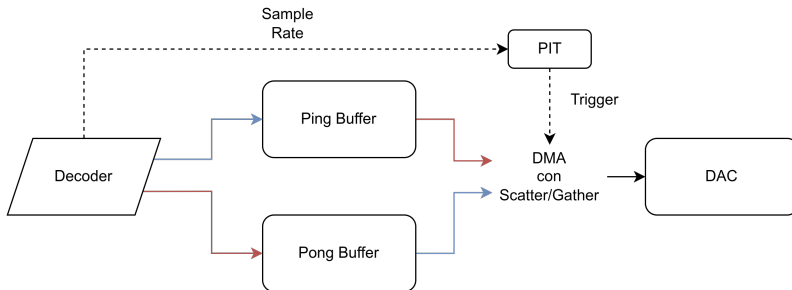
Uso de SDK



Uso de SDK: FATFS



Reproducción



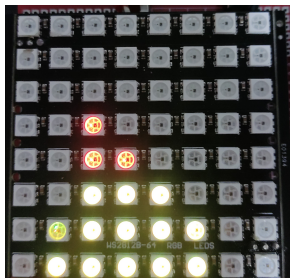
Matriz LED

```
typedef union
{
    uint32_t hex;
    struct
    {
        uint8_t b, g, r, bright;
    };
} color_t;
```

```
typedef struct
{
    color_t color;
    uint8_t onoff : 1;
    uint8_t blink : 1;
} LED_t;

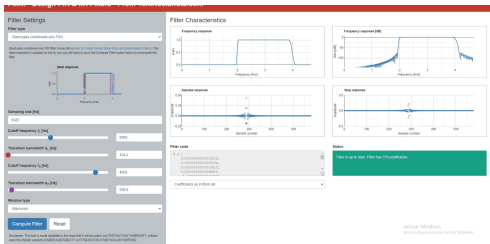
static LED_t LEDMatrix[NUMOFLEDS] = {0};
static uint16_t PWMLEDMatrix[NUMOFLEDS * RGBBITS + 2] = {0};
static uint8_t brightness = MAXBRIGHTNESS / 6;
static uint8_t refreshTimerID = 0;
static uint8_t blinkTimerID = 0;
```

Vúmetro



```
1 void initVumeter()
2 {
3
4     color_t aux;
5
6     initLEDMatrix();
7
8     for (int i = 0; i < COLS; i++)
9     {
10         for (int j = 0; j < ROWS; j++)
11         {
12             switch (j)
13             {
14                 case 0:
15                 case 1:
16                 case 2:
17                     aux.hex = YELLOW;
18                     changeColor(j, i, aux);
19                     break;
20
21                 case 3:
22                 case 4:
23                 case 5:
24                     aux.hex = RED;
25                     changeColor(j, i, aux);
26                     break;
27
28                 case 6:
29                 case 7:
30                     aux.hex = PURPLE;
31                     changeColor(j, i, aux);
32                     break;
33             }
34         }
35     }
36
37     adjustBrightness(2);
38     vumeterOn();
39 }
```

Ecualizador



$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

Accurate Cascade Graphic Equalizer

Vesa Välimäki, *Fellow, IEEE*, and Juho Liski

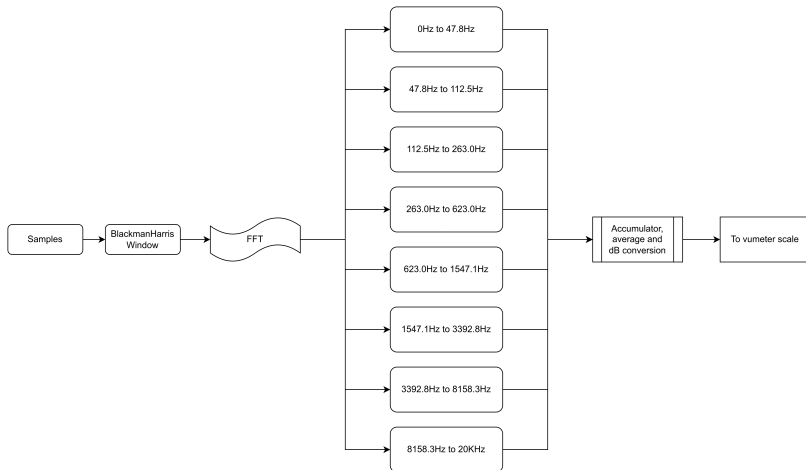
Abstract—A graphic equalizer is a high-order filter controlling the gain of several frequency bands. For good accuracy, graphic independent of each other, but this makes the overall cascade graphic EQ a very large filter [12], [13]. Some researchers

$$H_C(e^{j\omega T_s}) = G_0 \prod_{m=1}^M H_m(e^{j\omega T_s})$$

$$\mathbf{g}_{\text{opt}} = \mathbf{A}^{-1} \mathbf{t}$$

$$H(z) = \frac{1 + G\beta - 2\cos(\omega_c)z^{-1} + (1 - G\beta)z^{-2}}{1 + \beta - 2\cos(\omega_c)z^{-1} + (1 - \beta)z^{-2}}$$

Calculo de potencia por bandas



Libreria DSP



FIN

Gracias